PAGE NO.:
DATE: / /

UNIT-1

ASSIGNMENT-1

SECTION-A

Ques 1. What do you understand by DBMS?

A Database Management System (DBMS) is a software system that allows users to create, define, manipulate and manage databases. It provides a way for organizations to store, organize and retrieve large amounts of data quickly and efficiently in an organized manner.

Ques-2 What is E-R Model?

The Entity Relationship Model is a model for Identifying entities (like ~~stru~~ Student, car or company) to be represented in the database and representation of how those entities are related. The ER data model specifies enterprise schema that represents the overall logical structure of a database graphically.

Ques-3 Explain the importance of transaction Management.

Transaction mangement is crucial in databases to ensure data integrity and consistency during concurrent access and system failures. In a databases a transaction refers to a single logical operation on the data. It could be a simple read or write operation or a complex series of operations.

PAGE NO.:
DATE: / /

Ques-4 Explain the difference between data & Information.

Data comprises raw, unprocessed facts that need context to become useful, while information is data that has been processed, organized, and interpreted to add a meaning and value.

Ques-5 Describe the concept of KEYS in DBMS.

A key refers to an attribute / a set of attributes that helps us identify a row (or tuple) uniquely in a tuple (or relation). A key is also used when we want to establish relationships between the different columns and tables of a relational databases.

PAGE NO.:
DATE: / /

SECTION-B

QUES-1 What are the different parts of DBMS Languages? Also explain their commands.

Database languages are specialized languages used to interact with a database. Categorized into 4 types:

1. DDL (Data Definition Language)
- It is used to define the structure of database and pattern of the Database. It is used to define and modify the structure of the DB itself, including the tables, views, indexes and other schema related obj
- It deals with creation and modification of DB schema.
- DDL Commands:
  - CREATE
  - ALTER
  - DROP
  - TRUNCATE
  - RENAME

2. DCL (Data Control Language)

- DCL is used to access permissions of users to the DB. DCL commands help grant or revoke privileges to users, determining who can perform actions like reading or modifying data.
- DCL commands are transcational, meaning they can be rolled back if necessary.
- Grant: Gives user access to the databases.
- Revoke: Removes access or permissions from the user

3. DML (Data Manipulation Languge)

- It is used to manage and manipulate data within a DB.
- ~~Commands~~ The operations of the DML allow you to work with the actual content in the ~~DB~~ tables.
- DML Commands are as follows:
  - SELCT
  - INSERT
  - UPDATE
  - DELETE
  - MERGE
  - CALL
  - LOCK TABLE

4. TCL (Transaction Control Language)

- It is used to manage and control transactions in a DB, grouping them into logical units.
- These commands help to ensure the integrity of data and consistency during complex operations.
- there are two main commands of TCL.
  - Commit → used to save all changes made
  - Rollback → used to restore the DB to its original state from the last com

PAGE NO.:
DATE: / /

QUES-2 Define Entity, Attributes, Entity set, relationship with appropriate notations?

ENTITY

An Entity is a "thing" or "object" in the real world. An entity contains attributes, which describe that entity. So, anything about which we can store information is called an Entity. ~~En~~

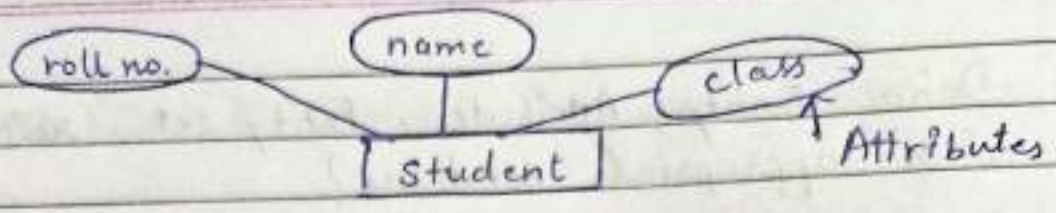
- Entities are recorded in the database and must be distinguishable, i.e easily recognized from the group.
- For Example, A student, An employee, or bank a/c all are entities.

~~A~~ [Entity] [WEAK ENTITY]

ATTRIBUTES

Attributes are properties or characteristics of an entity. Attributes are used to describe the entity. The attribute is nothing but a piece of data that gives more information about the entity.

- Attributes help to categorize the entity and the entity can be easily retrieved and manipulate the entity.
- Attributes can help the DB to be more structural and hierarchial.



3. ENTITY SET

An entity set is a collection of similar types of entities that share the same attributes.

- For example, All students of a school are a entity set of student entities.

- They can be classified into two types:

1. Strong Entity Set

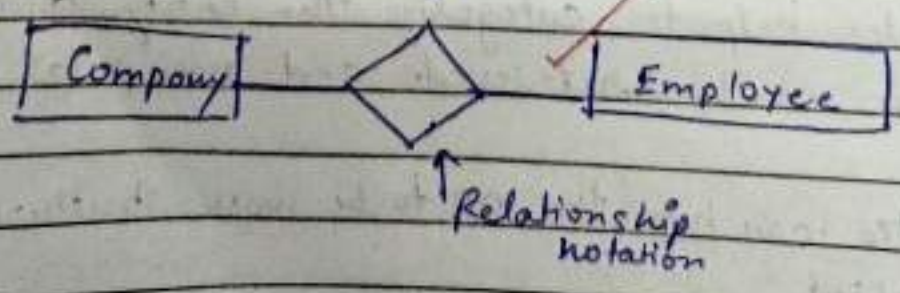
Strong Entity Set exists independently and each instance of a strong entity set has a unique primary key.

2. Weak Entity Set

A weak entity cannot exist on its own, it is dependent on a strong entity to identify it.

RELATIONSHIP

A relationship in a DBMS exists when the variable has a connection with the properties stored in different tables. Such relationships help the organization of entities intertwined with each other, ultimately enabling efficient data processing.



PAGE NO.:
DATE: / /

Ques-3 Explain data abstraction or 3 schema architecture.

Data Abstraction is a process of hiding unwanted or irrelevant details from the end user. It provides a different view and helps in achieving data independence which is used to enhance security of data.

Levels of abstraction in DBMS
Mainly there are 3 levels of abstraction:

1. PHYSICAL OR INTERNAL LEVEL
- It is the lowest level of abstraction for DBMS which defines how the data is actually stored, It defines data-structures to store data and access methods used by the database.

- The entire database is described in this level. It is a very complex level to understand. For example, customer's information is stored in tables and data is stored in the form of blocks of storage such as bytes, gigabytes etc.

2. LOGICAL OR CONCEPTUAL LEVEL
- Logical level is the intermediate level, It describes what data is stored in the DB and what relationship exists among those data. It tries to describe the entire or whole data because it describes what tables to be created and what are the links among those tables are created.

- It is less complex than physical level. logical level is

used by develops or database administrators (DBA)

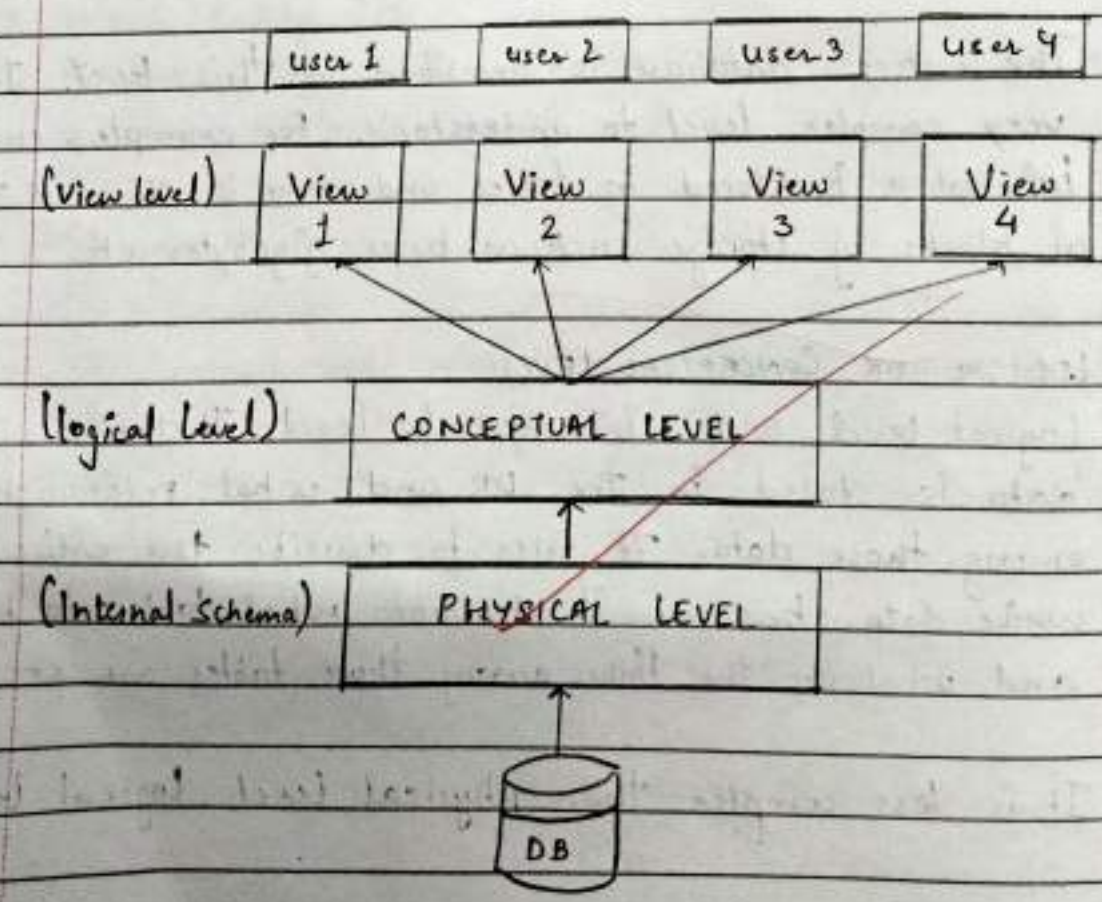
- The logical level contains tables (fields and attributes) and relationships among table attributes.

VIEW OR EXTERNAL LEVEL

- It is the highest level. In this level there are different levels of views and every view only defines a part of the entire data.

- It also simplifies interaction with the users and it provides many views or multiple views of the same DB.

- View level can be used by all users (all level's users). This level is the least complex and easy to understand.



PAGE NO :
DATE : / /

SECTION - C

QUES-1 What are DB Models on which DBMS is operated? Explain all in detail.

A Database model defines the logical design and structure of a database. It defines how data will be stored, accessed, and updated in a database management sys.

- As per your application's requirement, you can use a database model to define your database.
- You can define how you want to structure the application data using a database model.

→ Types of Database models

1. Hierarchical Model
- The hierarchical database model organizes data into a tree like structure, with a single root, to which all the other data is linked.

- The hierarchy starts from the root data, and expands like a tree, adding child nodes to the parent nodes.

- Data is organized into a tree like structure with a one-to-many relationship between two different types of data, for example, one department can have many courses, many teachers, and of course many students.

2. Network Model
- It is an Extention of Hierarchial Model.

PAGE NO.:
DATE: / /

- In this model, data is organized more like a graph, and allowed to have more than one parent node.

- Data is more related as more more relationships are established, as the more data is related, hence accessing the data is easier and is also fast.

3. Entity - relationship Model
- In this DBM, relationships are created by dividing ~~comp~~ objects of interest into entities and their characteristics into attributes.

- ER Models are defined to represent the relationships in pictorial form to make it easier for different stakeholders to understand.

4. Relational Model
- In this model, data is organized in two-dimensional tables and the relationship is maintained by storing a common field.

- This model ~~was intro~~ stores data in ~~tables~~ tabular structure. All the information related to a particular type stored in rows of that table.

5. Object Oriented Model
- In this model, data is stored in the form of objects

- The behavior of the object-oriented database model is just like object oriented programming.

- A very popular example of Object Database Management, is MongoDB which is also a NoSQL database.

6. NoSQL Model
- The NoSQL database model supports an unstructured style of storing data.
- Data is stored as documents.
- The documents look more like JSON strings or key value based object representation.
- It does provide features like indexing, relationships between data etc.

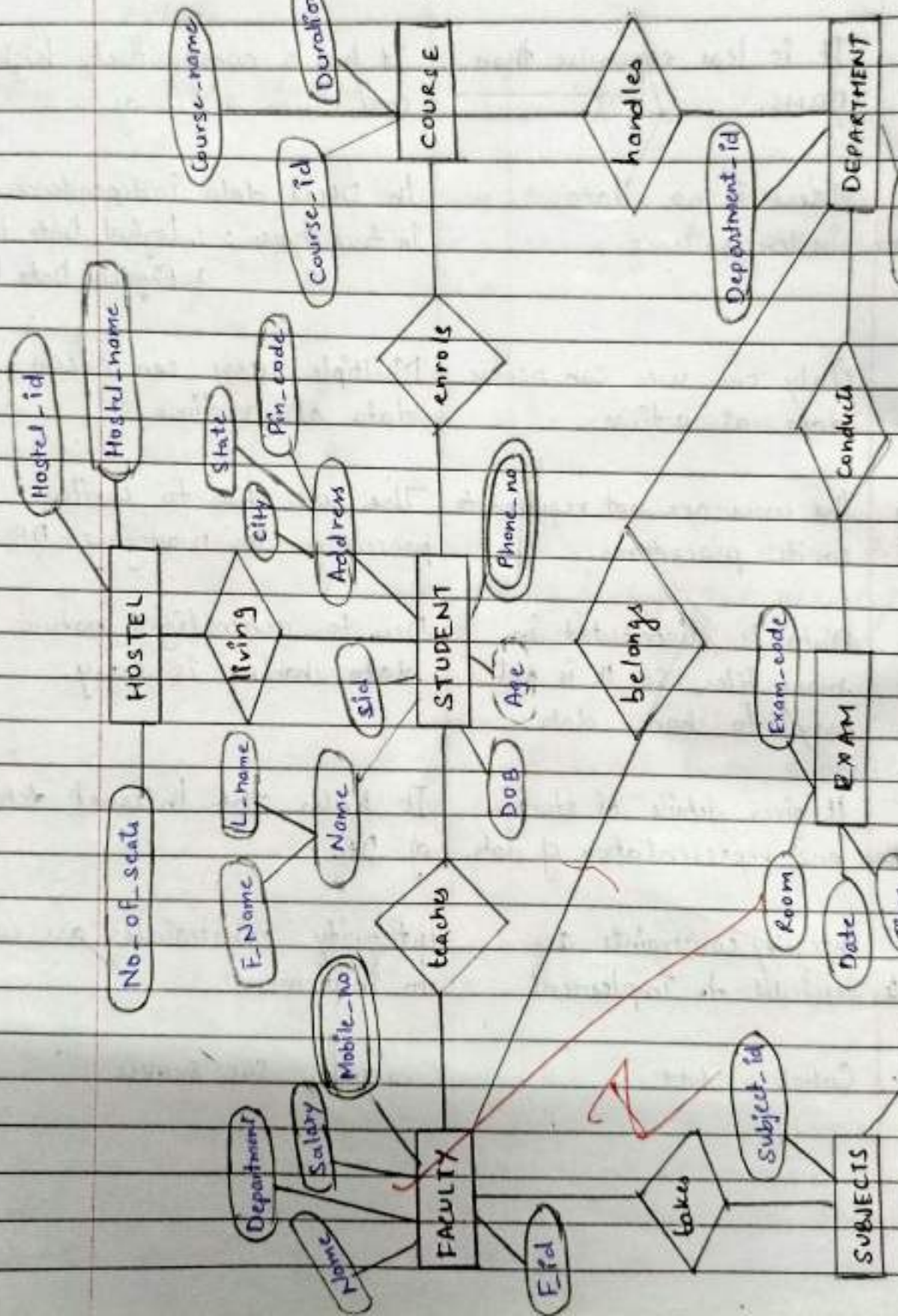
7. Graph Model
- The Graph database model is based on more real-world like relationships.
- Data representation using Nodes or entities.
- The nodes are related using edges.
- The popular DB Neo4J is based on the Graph DB model.
- In mordern apps like social networks, recommendation systems etc. the graph DB model is well suited.

PAGE NO.:
DATE: / /

Ques-2 Compare and Contrast file systems with database systems.

BASICS	FILE SYSTEM	DBMS
Structure	The file system is a way of arranging the files in a storage medium within a computer.	DBMS is software for managing the database.
Data Redundancy	Redundant data can be present in a file system	In DBMS is no redundant data.
Backup & Recovery	It doesn't provide inbuilt mechanism for backup and recovery of data if it is lost.	It provides in house tools for backup and recovery of data even if it is lost.
Query Processing	There is no efficient query processing.	Efficient query processing in DBMS.
Consistency	There is less data consistency in the file system.	There is more data consistency because of the process of normalization.
Complexity	It is less complex as compared to DBMS.	It has more complexity in handling
Security Constraints	File systems provide less security compared to DBMS	DBMS has more security mechanisms.

	FILE SYSTEM	DBMS
Cost	It is less expensive than DBMS.	It has a comparitively higher cost than a file system.
Data Independence	There is no data Indenpendence	In DBMS data Independence exists in two types : 1. Logical Data Indep. 2. Physical Data Indep.
User Access	Only one user can access data at a time.	Multiple users can access data at a time.
Meaning	The users are not required to write procedures	The user has to write procedures for managing DB.
Sharing	Data is distributed in many files. So, It is not easy to share data.	Due to centralized nature data sharing is easy.
Data Abstraction	It gives details of storage and representation of data.	It hides the Internal details of DB.
Integrity Constraints	Integrity constraints are difficult to implement	Integrity constraints are easy to implement.
Example	Cobol , C++ ,	Oracle , SQL Server

PAGE NO.:
DATE: / /

Ques - 3 Draw the E-R diagram of University Management Syst